

Introduction

Wound Care

- Almost **5 million** people die from **insufficient wound care** annually (Mujawar et al., 2025).
- The wound healing process benefits from **moist conditions**, which allow for cell signaling, and a **mechanical barrier** over the wound to prevent infection (Zhang & Zhao, 2020).
- The ineffectiveness of **common wound dressings** (foams, films, gauzes) can be attributed to their **lack of absorbency and insufficient mechanical strength**.
- Hydrogels and the **sodium alginate-gelatin (GELSA)** polymer blend in particular have been observed to significantly **accelerate wound closure** and reduce scarring (Mujawar et al., 2025).

Galleria mellonella

- The greater wax moth is often used to model injuries and infections due to its **affordability** and **comparable immune system** to that of a human (Maslova et al., 2023).
- *G. mellonella* is widely considered to be an **ethical** organism to experiment on.

Purpose

Compare the rate of wound closure in organisms treated with GELSA hydrogel to that of organisms treated with a commercial liquid bandage.

Hypothesis

The rate of wound closure in organisms treated with GELSA hydrogel will be more rapid than that of those treated with New Skin brand liquid bandages because GELSA hydrogel possesses absorptive properties lacked by film dressings and contains gelatin, a polypeptide that promotes the proliferation of collagen.

Experimental Design

Research Question

How does the rate of wound closure in burned *G. mellonella* larvae treated with GELSA hydrogel compare to that of larvae treated with New Skin liquid bandages?

IV: Type of wound treatment

- GELSA hydrogel (GELSA)
- New Skin liquid bandages (NLB)
- No treatment (NONE)

DV: Rate of wound closure in mm^2 per day

+ Control: Larvae treated with New Skin, expected to have higher rates of wound closure than untreated larvae

- Control: Larvae left untreated, expected to have lowest rates of wound closure

Comparing the effect of sodium alginate-gelatin hydrogel dressings on the rate of wound closure in burned *G. mellonella* larvae to that of liquid bandages

Methodology

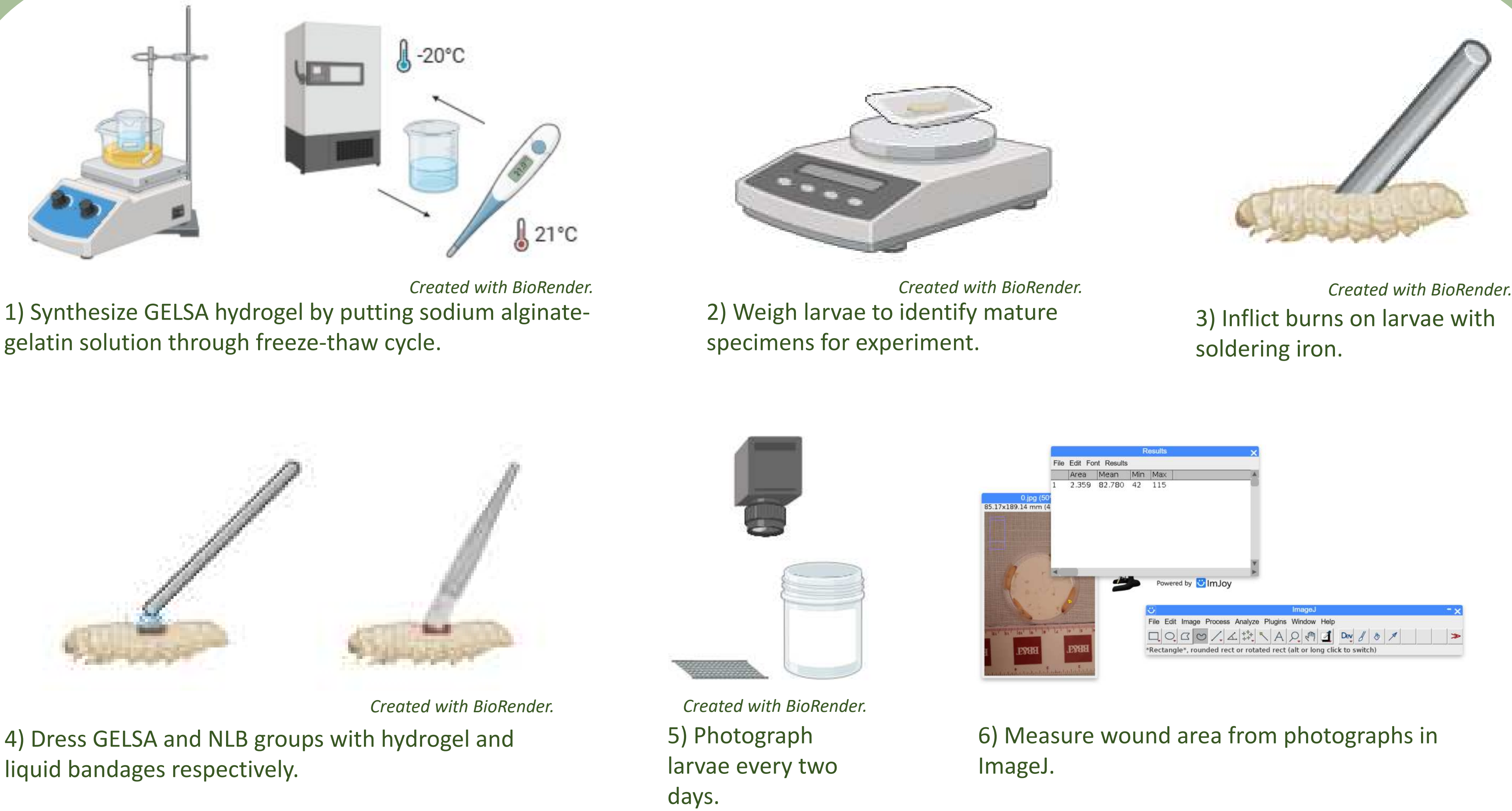


Figure 1: Visualization of experimental methods. Created by researcher in BioRender. <https://www.biorender.com/>

Data and Results

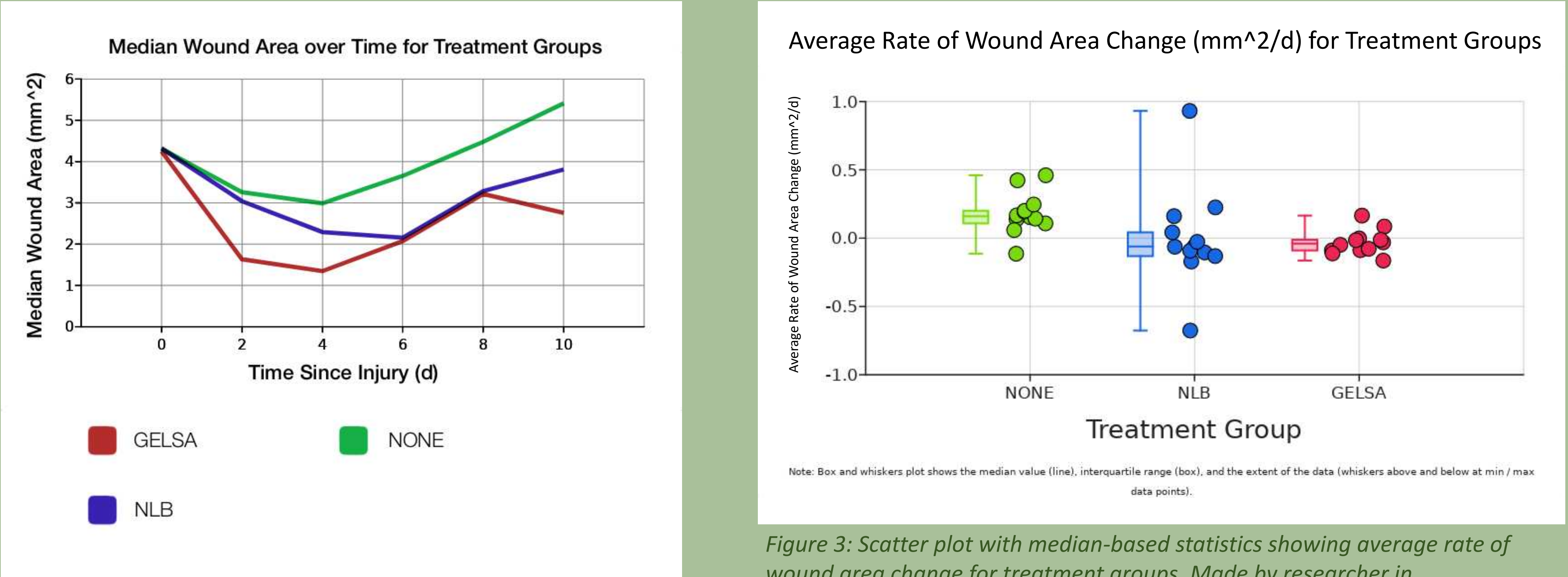


Figure 2: Graph showing median wound area over time for treatment groups. Made by researcher in DataClassroom. <https://dataclassroom.com/en>

Figure 3: Scatter plot with median-based statistics showing average rate of wound area change for treatment groups. Made by researcher in DataClassroom. <https://dataclassroom.com/en>

Median Wound Area over Time for Treatment Groups							
Median Wound Area (mm^2)	Treatment Group	Time (d)					
		0	2	4	6	8	10
	GELSA	4.253	1.634	1.345	2.076	3.215	2.757
	NLB	4.321	3.041	2.292	2.154	3.286	3.809
	NONE	4.314	3.258	2.991	3.654	4.480	5.413

Figure 4: Table of median wound area over time for treatment groups. Made by researcher in Google Sheets. <https://docs.google.com/spreadsheets/u/0/>

Stats

$\alpha = 0.05$ $df = 2$ $n = 12$

H_0 = The median average rate of wound closure is the **same** between treatment groups.
 H_A = The median average rate of wound closure is **not the same** between treatment groups.

Because the H-statistic of 9.960 is greater than the critical value of 5.872 and the p-value of 0.01 is less than the significance level of 0.05, we **reject the null hypothesis** that the median average rate of wound closure is the same between treatment groups. Therefore, the **median average rate of wound closure** is **not the same** between treatment groups. A post-hoc Dunn test reveals that there is a p-value of 0.02 between NONE and NLB groups and between NONE and GELSA groups. However, the p-value between NLB and GELSA groups was 1.00. Therefore, while there is strong evidence that both NLB and GELSA are **different from NONE**, the **NLB and GELSA** groups were **not significantly different**.

Conclusions

While the **liquid bandage and GELSA** hydrogel treatments **both significantly accelerated** the rate of wound closure in burned *G. mellonella*, there was **no evidence that one was more conducive** to healing than the other. However, data in the NLB group appeared to be more affected by **outliers** due to the relatively large standard error, so further experimentation with **larger cohorts** could yield more accurate results. Additionally, testing the **mechanical properties** of New Skin and GELSA hydrogel could be used to explain their wound healing expediting abilities comparative to one another. Despite the results unexpectedly suggesting that GELSA hydrogel and New Skin liquid bandage were equally effective in accelerating wound closure, the **significantly lower rate** of wound closure in the **untreated** (negative control) group **justifies *Galleria mellonella* as a model of wound healing** because their burns responded to a lack of treatment as a human's would. **29 of 30 larvae also survived** through the process of data collection, further proving their ability to be used as **healing models** rather than subjects of a mortality assay.

References

Maslova, E., Osman, S., & McCarthy, R. R. (2023, June 1). Using the *Galleria mellonella* burn wound and infection model to identify and characterize potential wound probiotics. *Microbiology*, 169(6). Europe PMC. <https://doi.org/10.1099/mic.0.001350>

Mujawar, S. S., Arbade, G. K., Bisht, N., Mane, M., Tripathi, V., Sharma, R. K., & Kashte, S. B. (2025, January 15). 3D printed Aloe barbadensis loaded alginate-gelatin hydrogel for wound healing and scar reduction: In vitro and in vivo study. *International Journal of Biological Macromolecules*, 296. ScienceDirect. <https://doi.org/10.1016/j.ijbiomac.2025.139745>

Zhang, M., & Zhao, X. (2020, August 13). Alginate hydrogel dressings for advanced wound management. *International Journal of Biological Macromolecules*, 162, 1414-1428. ScienceDirect. <https://doi.org/10.1016/j.ijbiomac.2020.07.311>